

CS 1671 / CS 2071

Human Language Technologies

Session 9: Project match day, CRCD tutorial

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University of
Pittsburgh

School of Computing and Information

Quiz

- Go to **Quizzes > Quiz 09-22** on Canvas.
 - Session 7, N-gram language models: J+M 3-3.6.2, 3.8
 - Session 8, Logistic regression part 1: J+M 4-4.3
- You have until **1:10pm** to complete it
- Allowed resources
 - Textbook
 - Your notes (on a computer or physical)
 - Course slides and website
- Resources not allowed
 - Generative AI
 - Internet searches

Assessments: project

After your project group is formed today:

- Establish a communication channel (Discord, Teams through Pitt, email, Signal, WhatsApp, etc)
- [Project deliverable 1](#), due Oct 8, is the next project assignment
 - Michael may update some of these instructions, but the gist will be similar

Overview: Project match day

- Project match process
- CRCDD resources available for the project

Project match

- Go to the spot in the room with the project paper you are most interested in working on
 - We will do this for multiple rounds
- **Goal: groups of ~5 on projects**
- After groups are finalized, write your names on the back of the project paper

CRCDD resources for the project

CRCO resources available for the project

- Storage space
 - 5 TB shared space for the whole class at /ix1/cs1671-2026f
- Run scripts on the command line through the SLURM job scheduler
 - Tons of CPUs and powerful GPUs available
 - <https://crc.pitt.edu/overview-crc-services/computing-hardware>
- Jupyter notebooks
 - Teach cluster
 - OnDemand

Logging into the CRCD with CLI

`ssh <Pitt username>@h2p.crc.pitt.edu`

- You will need to be on the [Pitt VPN](#) ([GlobalProtect app](#)) if you are not connected to WIRELESS-PITTNET
- Your home directory only has 75 GB of storage!
- Check quota use with `crc-quota` command
- Feel free to store project data, code, etc at class storage space `/ix1/cs1671-2026f`
 - 5 TB available



Photo: Chuck Stout, from NARA & DVIDS

Creating a Python environment on the CRCSD for your project

Managing Python environments on the CRCD

2 recommended options to creating a Python environment for your project:

1. Copy the class environment
2. Create a new environment for your project

Please don't use the actual class environment for your project, since that can slow things down (and you shouldn't install new packages there)

Install Pixi within your CRCD user directory

```
curl -fsSL https://pixi.sh/install.sh | sh
```



Option 1: Copying the class environment for your project

1. Create a new directory for your project in `/ix1/cs1671-2026f`
`mkdir <project-name>`
2. Copy `pyproject.toml`, the manifest file with all the packages
`cp class-env/pyproject.toml <project-name>`
3. Edit `pyproject.toml` in a terminal text editor (nano, vim, emacs, etc)
 - Change the name field from `class-env` to your *project-name*
 - Feel free to delete any packages you don't think you'll use (you can always add them back later)
4. Set your Pixi cache inside `/ix1/cs1671-2026f`
`export PIXI_CACHE_DIR="/ix1/cs1671-2026f/.<project-name>_pixi_cache"`
5. Run `pixi install` (will probably take a long time)

Option 2: Create a new environment for your project

1. Create a new directory for your project in `/ix1/cs1671-2026f`
 - `mkdir project-name`
2. Create a new pixi environment
 - `pixi init project --format pyproject`
 - `pixi add <package> --pypi`
 - For pip packages
- Add ipykernel and jupyterlab for it to work with loading Jupyter notebooks on the teach cluster
 - Talk to Michael if it doesn't work for you

Jupyter options on the CRCd

Jupyter options on CRCD: teach cluster

JupyterHub on the teach cluster that we've been using in class is fine to use

- Please load your own project Python environment as a custom path, not the class environment
- Feel free to select GPU options if needed



Jupyter options on CRCSD: Open OnDemand

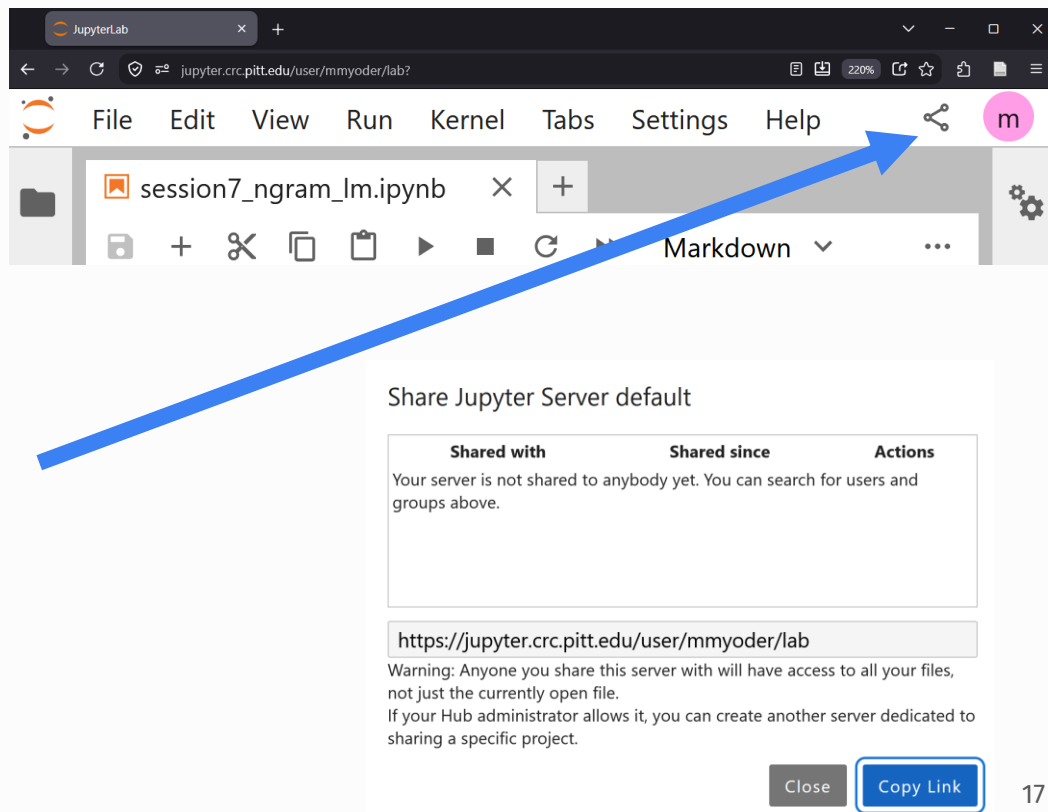
If you need something for longer than 3 hours, see documentation on Open OnDemand (which also has an R portal): <https://crc-pages.pitt.edu/user-manual/web-portals/jupyter-ondemand/>



- You request a server and they notify you when it's available
- You can provide a path to a custom Python environment
- Email Michael if you can't log in or have other issues

Collaborating on Jupyter notebooks

- Collaboration option available on the teach cluster (not Open OnDemand)
- Click share icon
- Click copy link to share document links with others
- jupyter-collaboration extension [documentation](#)
- Can use external services like Google Colab, though you might run up against resource constraints



The screenshot shows the JupyterLab web interface in a browser. The address bar displays `jupyter.crc.pitt.edu/user/mmyoder/lab?`. The top menu bar includes File, Edit, View, Run, Kernel, Tabs, Settings, and Help. A blue arrow points from the 'Help' menu to a share icon (a network of nodes) located next to a user profile icon labeled 'm'. Below the menu bar, the active notebook is `session7_ngram_lm.ipynb`. The toolbar shows various icons for file operations and execution. A sharing dialog is open, titled 'Share Jupyter Server default'. It contains a table with columns 'Shared with', 'Shared since', and 'Actions'. The table body shows a warning: 'Your server is not shared to anybody yet. You can search for users and groups above.' Below the table, the shareable URL is displayed: `https://jupyter.crc.pitt.edu/user/mmyoder/lab`. A warning message follows: 'Warning: Anyone you share this server with will have access to all your files, not just the currently open file. If your Hub administrator allows it, you can create another server dedicated to sharing a specific project.' At the bottom right of the dialog are 'Close' and 'Copy Link' buttons.

Shared with	Shared since	Actions
Your server is not shared to anybody yet. You can search for users and groups above.		

<https://jupyter.crc.pitt.edu/user/mmyoder/lab>

Warning: Anyone you share this server with will have access to all your files, not just the currently open file. If your Hub administrator allows it, you can create another server dedicated to sharing a specific project.

Close Copy Link

Running command-line scripts on the CRCD

Running scripts with SLURM job scheduler

- You can run scripts (like Python scripts) on the CRCD, just **don't run them directly on the nodes that you log into with ssh**
- Write a shell script with the commands you want and SLURM options at the top
- See the CRCD documentation: <https://crc-pages.pitt.edu/user-manual/slurm/batch-jobs/>



Using your project Python environment in SLURM

In shell scripts for SLURM, run Python files with pixi run

```
cd <your_project_directory>
```

```
pixi run python <python_script.py>
```